Лабораторная работа №1

Установка ОС

Юсупова Амина Руслановна

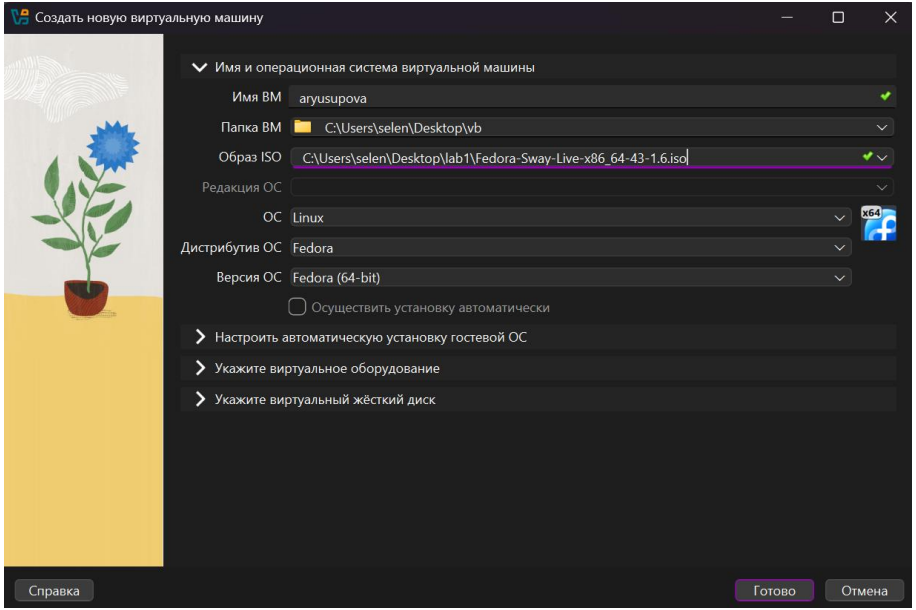
ЦЕЛИ И ЗАДАЧИ РАБОТЫ

Цель лабораторной работы

Целью данной работы является приобретение практических навыков установки операционной системы на виртуальную машину, настройки минимально необходимых для дальнейшей работы сервисов

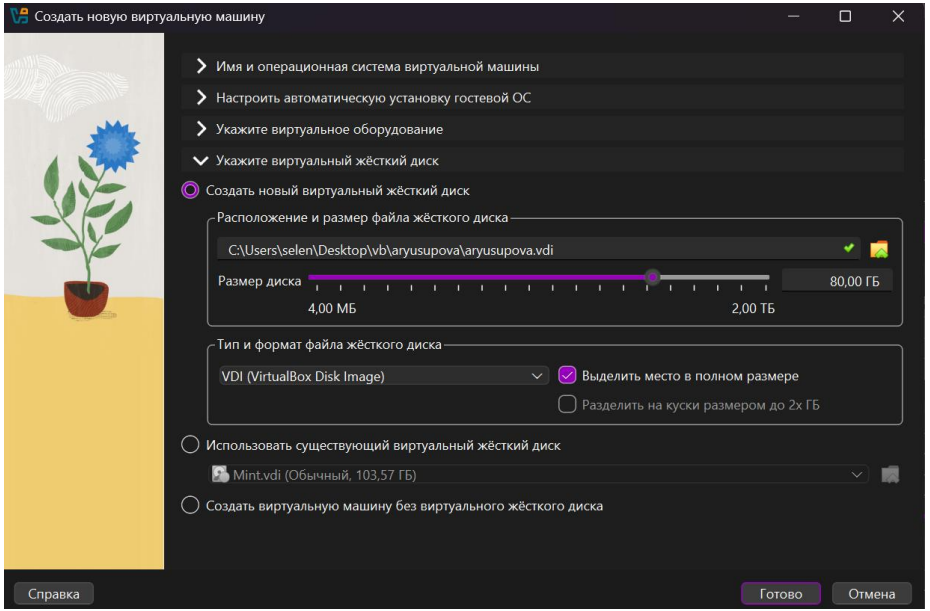
ПРОЦЕСС ВЫПОЛНЕНИЯ ЛАБОРАТОРНОЙ РАБОТЫ

Создаю виртуальную машину



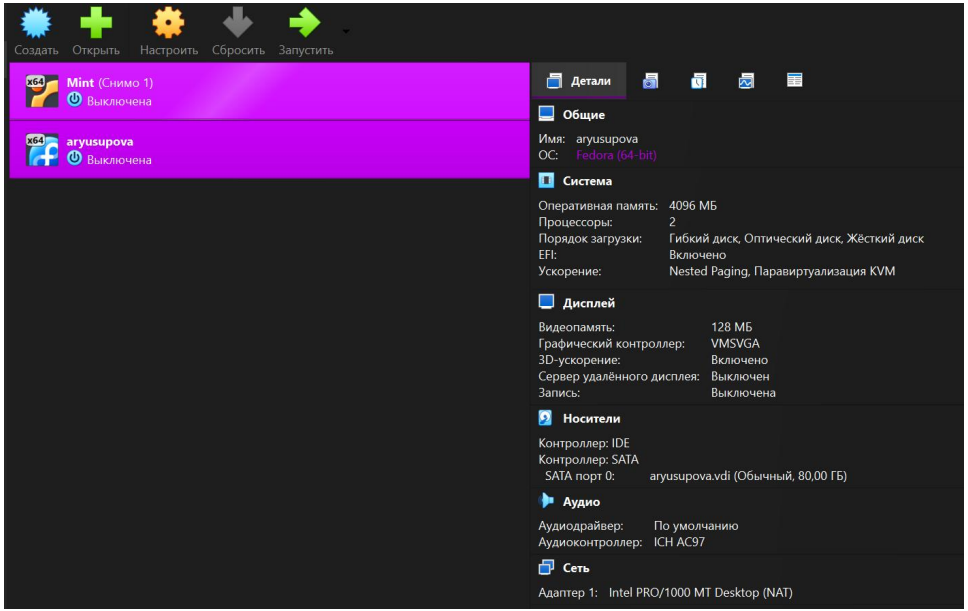
Создание новой виртуальной машины

Задаю конфигурацию жёсткого диска



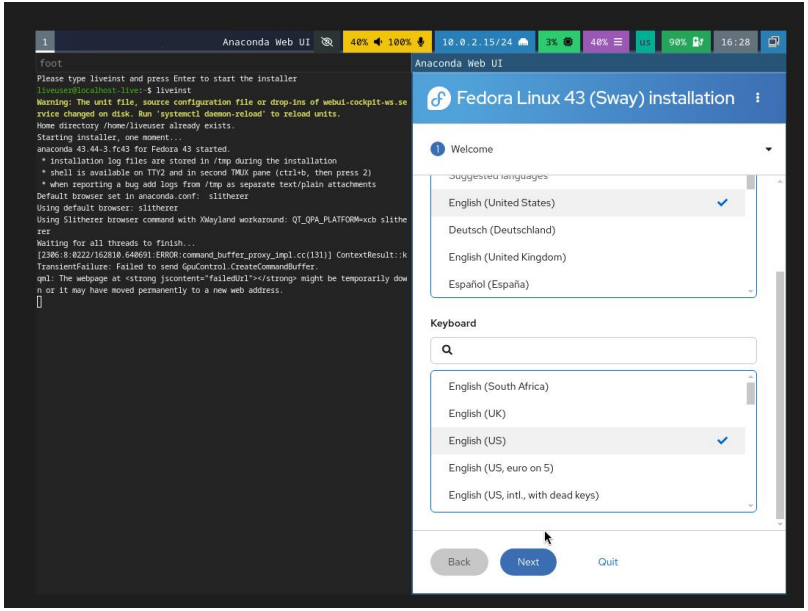
Конфигурация жёсткого диска

Добавляю новый привод оптических дисков и выбираю образ



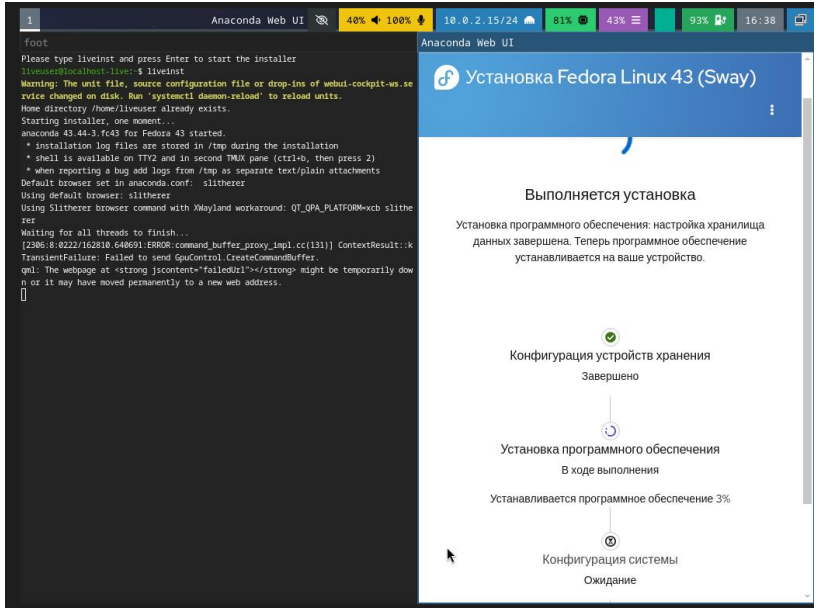
Конфигурация системы

Установка языка



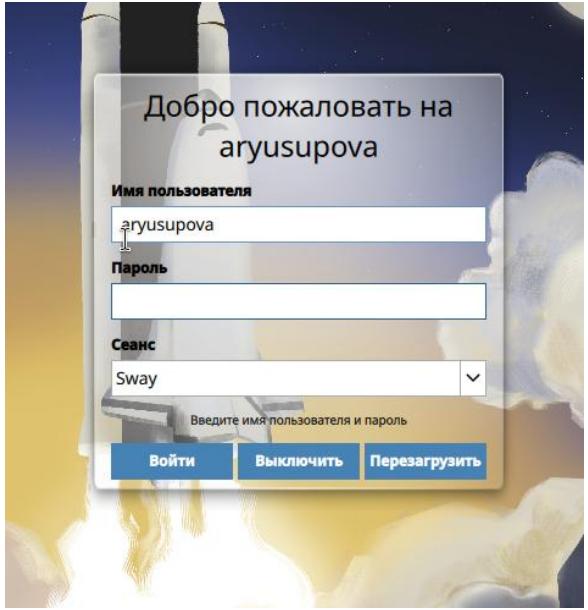
Установка языка

Завершение установки



Завершение установки

Создание пользователя и вход в учётную запись



Вход в учетную запись

Рабочая система

```
root@aryusupova:~# dmesg | grep 'Linux ver'
[    0.000000] Linux version 6.17.1-300.fc43.x86_64 (mockbuild@5381c258a4b3436489a448ea66bda8ce) (gcc (GCC) 15.2.1 20250924 (Red Hat 15.2.1-2), GNU ld version 2.45-1.fc43
) #1 SMP PREEMPT_DYNAMIC Mon Oct  6 15:37:21 UTC 2025
root@aryusupova:~# dmesg | grep MHz
[    0.000024] tsc: Detected 1497.603 MHz processor
[   19.797272] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:39:04:1c
root@aryusupova:~# dmesg | grep 'cpu0'
root@aryusupova:~# dmesg | grep -i 'cpu0'
[    1.381997] smpboot: CPU0: Intel(R) Core(TM) i5-1035G4 CPU @ 1.10GHz (family: 0x6, model: 0x7e, stepping: 0x5)
root@aryusupova:~# dmesg | grep Mem
[    0.000000] DMI: Memory slots populated: 0/0
[    1.410440] Memory: 3919852K/4174724K available (22017K kernel code, 4533K rwdata, 17980K rodata, 5116K init, 6152K bss, 245032K reserved, 0K cma-reserved)
[    1.414800] x86/mm: Memory block size: 128MB
[    4.973627] systemd[1]: memstrack.service - Memstrack Anylazing Service was skipped because no trigger condition checks were met.
root@aryusupova:~# dmesg | grep Hyper
[    0.000000] Hypervisor detected: KVM
root@aryusupova:~# dmesg | grep root
[    0.000000] Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.17.1-300.fc43.x86_64 root=UUID=114b3d1d-5be0-4ff7-b316-a8458e118a37 ro rootflags=subvol=root nomodeset rhgb q
uiet
[    0.531881] Kernel command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.17.1-300.fc43.x86_64 root=UUID=114b3d1d-5be0-4ff7-b316-a8458e118a37 ro rootflags=subvol=root nomodeset
 rhgb quiet
[    1.550723] pci_bus 0000:00: root bus resource [io  0x0000-0x0cf7 window]
[    1.550729] pci_bus 0000:00: root bus resource [io  0x0d00-0xffff window]
[    1.550732] pci_bus 0000:00: root bus resource [mem 0x000a0000-0x000bffff window]
[    1.550735] pci_bus 0000:00: root bus resource [mem 0xe0000000-0xfdffffff window]
[    1.550738] pci_bus 0000:00: root bus resource [bus 00-ff]
[    1.839386] Trying to unpack rootfs image as initramfs...
[   12.611691] systemd[1]: initrd-switch-root.service: Deactivated successfully.
[   12.612478] systemd[1]: Stopped initrd-switch-root.service - Switch Root.
[   12.632777] systemd[1]: Stopped target initrd-switch-root.target - Switch Root.
[   12.633103] systemd[1]: Stopped target initrd-root-fs.target - Initrd Root File System.
[   12.797706] systemd[1]: plymouth-switch-root.service: Deactivated successfully.
[   12.803582] systemd[1]: Stopped plymouth-switch-root.service - Plymouth switch root service.
[   12.805670] systemd[1]: systemd-fsck-root.service: Deactivated successfully.
[   12.805754] systemd[1]: Stopped systemd-fsck-root.service - File System Check on Root Device.
root@aryusupova:~# dmesg | grep -i 'mount'
[0] 0:sudo*                                                                    "aryusupova" 01:09 24-фев-26
```

Команда dmesg

Рабочая система

```
tmpfs                397728           60   397668             1% /run/user/0
root@aryusupova:~# dmesg | grep root
[    0.000000] Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.17.1-300.fc43.x86_64 root=UUID=114b3d1d-5be0-4ff7-b316-a8458e118a37 ro rootflags=subvol=root nomodeset r
uiet
[    0.531881] Kernel command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.17.1-300.fc43.x86_64 root=UUID=114b3d1d-5be0-4ff7-b316-a8458e118a37 ro rootflags=subvol=root nomo
 rhgb quiet
[    1.550723] pci_bus 0000:00: root bus resource [io  0x0000-0x0cf7 window]
[    1.550729] pci_bus 0000:00: root bus resource [io  0x0d00-0xffff window]
[    1.550732] pci_bus 0000:00: root bus resource [mem 0x000a0000-0x000bffff window]
[    1.550735] pci_bus 0000:00: root bus resource [mem 0xe0000000-0xfdffffff window]
[    1.550738] pci_bus 0000:00: root bus resource [bus 00-ff]
[    1.839386] Trying to unpack rootfs image as initramfs...
[   12.611691] systemd[1]: initrd-switch-root.service: Deactivated successfully.
[   12.612478] systemd[1]: Stopped initrd-switch-root.service - Switch Root.
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[   12.633103] systemd[1]: Stopped target initrd-root-fs.target - Initrd Root File System.
[   12.797706] systemd[1]: plymouth-switch-root.service: Deactivated successfully.
[   12.803582] systemd[1]: Stopped plymouth-switch-root.service - Plymouth switch root service.
[   12.805670] systemd[1]: systemd-fsck-root.service: Deactivated successfully.
[   12.805754] systemd[1]: Stopped systemd-fsck-root.service - File System Check on Root Device.
root@aryusupova:~# df
Файловая система 1K-блоков Использовано Доступно Использовано% Смонтировано в
/dev/sda3         81172480     10371784 69693448           13% /
devtmpfs           1964844            0  1964844            0% /dev
tmpfs              1988644         3420  1985224            1% /dev/shm
efivarfs               256           38      214           15% /sys/firmware/efi/efivars
tmpfs               795460         1188   794272            1% /run
tmpfs                 1024            0     1024            0% /run/credentials/systemd-journald.service
tmpfs              1988644            4  1988640            1% /tmp
/dev/sda3         81172480     10371784 69693448           13% /home
/dev/sda2         1992552       353452  1517860           19% /boot
/dev/sda1          613160        19764   593396            4% /boot/efi
tmpfs                 1024            0     1024            0% /run/credentials/systemd-resolved.service
tmpfs               397728           88   397640            1% /run/user/1000
tmpfs               397728           60   397668            1% /run/user/0
root@aryusupova:~#
```

Команда dmesg

ВЫВОДЫ ПО ПРОДЕЛАННОЙ РАБОТЕ

Вывод

Мы приобрели практические навыки установки операционной системы на виртуальную машину, настройки минимально необходимых для дальнейшей работы сервисов.